

TRIAL OF THE PROPHYLAXIS OF BOVINE TUBERCULOSIS BY BCG IN AN INFECTED RURAL HOLDING

Gruville, Seine-Inférieure – 1921 to 1927

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ABOUT THIS TRANSLATION

Translated in full from the printed French text of the Gallica digitisation of the Institut Pasteur copy (ark:/12148/bpt6k5791139p, views f32 to f52), working from the page images rather than the machine OCR, which corrupts exactly the material that matters most here: animal numbers, dates, weights and place names.

CONVENTIONS

Original page numbers are marked in the margin of the text. Paragraph breaks, headings, italics and running order follow the original. Terms of art are rendered literally: *tuberculation* as tuberculin testing, *saisie* as condemnation, *tuerie particulière* as private slaughterhouse, *réagissant* as reactor. Translator's notes are marked **TN** and are the only text here that is not in the original.

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The estate of Gruville, belonging to M. Eugène Le Grand, general councillor of the canton of Valmont, occupies, 8 kilometres from Fécamp (Seine-Inférieure), an area of 80 hectares, of which 40 are in grassland and 40 under cultivation. Cattle are raised there for butter production and for fattening.

In general, apart from cases of malformation, all the calves are reared. Males and females are bucket-fed during their first month with the mother's milk, unboiled, given in three feeds. During the second month they receive, in two feeds, a mixture in equal parts of whole milk and skimmed milk from the dairy. From the third to the fourth month the milk in this mixture is progressively replaced by a "drink" [*buvée*] of linseed cake. This mixture of skimmed milk and cake is given until the age of seven to eight months as a supplement to the grass ration, which begins in the second month, the calves being then turned out loose to pasture and able to shelter day and night in an open byre.

Males and females over one year old are permanently on the open-air regime, at grass, receiving in winter a ration of sugar-beet pulp. They may come in at night to open byres serving as shelter. Fattened bullocks are

sent to the abattoir at the age of thirty to forty months. The dairy cows are fed on chopped fodder beet, linseed cake, hay and straw. From November to April the stock is distributed among five byres of modern construction, well ventilated and kept in perfect cleanliness.

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TUBERCULOUS INFECTION ON THE ESTATE

The first administrative document officially establishing the existence of tuberculosis at Gruville dates from 28 June 1915: a bullock of twenty-seven months, born on the farm and sent to the public abattoirs of Fécamp, was found there to be carrying numerous tubercles on the lungs, the heart, the pericardium, the costal pleura, and in some mesenteric lymph nodes. Sub-caudal intradermal tuberculin testing, carried out on 31 July 1915 as a consequence of the declaration transmitted by the Fécamp abattoirs, gave, among the 67 cattle then making up the Gruville herd, 32 reactions (26 very clear and 6 suspect), that is 47.7 per 100. Among these 32 infected subjects there were 15 cows out of 19 tested, 7 heifers out of 20, and 10 calves and bullocks from nine months to three years out of 28 tested.

All these animals were, in accordance with article 46 of the decree of 6 October 1904, placed under the supervision of the departmental Epizootics Service, and their isolation was carried out so far as was possible. But the observations subsequently made on the other cattle which had not reacted supplied proof that these isolation measures had not sufficed to preserve them from contagion; so that a new prefectoral order had to be issued, on 16 September 1918, for the whole cattle holding, which then comprised 69 *head*, of which 18 dairy cows, 19 heifers, 16 bullocks of one to two years and 16 calves of fifteen days to seven months.

In 1919, M. Eugène Le Grand decided to have all his cattle tuberculin-tested. After a sub-caudal intradermal test made on 5 April, and an ophthalmic test with crude tuberculin on 26 June, the order of 16 September 1918, partially lifted, released 36 non-reacting animals, leaving 26 reactors under sanitary supervision (12 cows, 13 heifers and 1 bullock of fifteen months), whose progressive elimination the owner resolved to pursue in order to free his holding. Consignments to the abattoir were soon staged under the conditions required by the sanitary regulations and, by the end of 1920, that is to say within eighteen months, 14 reactors had been slaughtered.

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Sanitary inspection had found in these animals two cases of massive tuberculosis warranting total condemnation; one tuberculosis localised to the right lung and the right costal pleura entailing condemnation of a forequarter, of 60 kilogrammes; three cases of pulmonary tuberculosis; one of hepatic tuberculosis. In the seven other animals no lesion was discovered.

Thus, at the end of 1920, tuberculous infection remained manifest at the Gruville farm, which still counted at that date 12 reacting cattle out of the 26 denounced by the tuberculin testing of 1919, which had revealed, in this herd of 62 head, a proportion of 41.7 *per 100 infected subjects*.

It was at this period, November 1920, following the publication in the *Annales de l'Institut Pasteur* (No. 9, September 1920) of a memoir by M. Calmette in collaboration with one of us, that the possibility arose of carrying over into agricultural practice the vaccination trials with BCG, until then confined to laboratory experiments. Informed of the interest of this research and of the guarantees it offered as to the harmlessness of the bile-grown bacillary cultures, M. Eugène Le Grand, for reasons in which scientific interest counted for as much as altruistic sentiment, hastened to place his cattle herd at our disposal with a view to the trials envisaged.

The problem we set ourselves to resolve was the following:

In a holding infected with tuberculosis, without changing anything whatever in the manner of life or the housing of the animals, without modifying the usual methods of rearing the young, is it possible, through the normal course of births, by vaccinating the newborn in the first fifteen days of their life, and by revaccinating them each year, to purge this holding of tuberculosis within a period of five years?

It was decided that the vaccination trials would begin at Gruville from 1 January 1921, *on all calves to be born after that date*; that no controls would be kept among the young subjects, on account of the size of the percentage of contagion already achieved, a percentage which the trials were to reduce or abolish if the method proved effective. The

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numbers under experiment were therefore to increase from year to year, following the rhythm of the births, and the vaccinated and revaccinated animals were to be kept until the end of their normal use under the ordinary conditions of regional practice.

Furthermore, in agreement with the departmental Veterinary Service, *all the prophylactic measures of isolation and disinfection provided for by the regulations were suspended*.

II

CAMPAIGN OF 1921

In the course of this year we had conceived, knowing nothing of the trials carried out by H. Vallée in a somewhat different sense (but proceeding from the same idea) and which he has published,¹ the notion of creating, under the skin of the animals, a *persistent and active local lesion* by inserting into the subcutaneous

¹ Acad. des Sc., 2 January 1924.

connective tissue of the dewlap of the calves, from the first month of their life, a *ball of pumice stone* of 10 millimetres, charged, under vacuum, with a concentrated emulsion of vaccine bacilli.

There were, in this year 1921, 22 births (8 males and 14 females). *All these calves were vaccinated by subcutaneous insertion of a vaccine pellet.*

It had been foreseen at that time that vaccination should be carried out about one month after birth. Owing to the sequence of calvings and to the convenience of travel (one of us coming from Lille to set the method going), the vaccination operations followed one another as below: 17 February (2), 3 March (3), 15 March (5), 20 April (4), 31 May (2), 5 July (2), 20 October (2), 6 December (2).

These operations are distributed as follows with respect to the age of the calves on the day of the intervention:

AGE AT VACCINATION – 1921 GENERATION

1ST WEEK	2ND WEEK	3RD WEEK	4TH WEEK	6TH WEEK	8TH WEEK
2	5	8	5	1	1

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The male calf no. 49, born 31 August 1921, having been able to be operated on only on 20 October, on the fiftieth day, had beforehand been tested by the ophthalmo-reaction with crude tuberculin: negative result on 15 October.

MORTALITY DURING THE YEAR AMONG THE VACCINATED, FROM CAUSES OTHER THAN TUBERCULOSIS

Four vaccinated calves (1 male and 3 females) died in 1921: a female of one month, of infectious broncho-pneumonia; a male of three months (the no. 49 mentioned above) of verminous bronchitis; a female of three months of suppurative sinusitis; and a female of five months (*Orion*, no. 55, born 28 May and vaccinated on the 31st, on the third day), which died on 22 November 1921 of acute peritonitis of traumatic origin. An autopsy having been possible, lymph nodes were taken from the thoracic and abdominal cavities, as well as the vaccine pellet inserted under the skin of the dewlap, which locally had provoked no reaction. Inoculation of these nodes into guinea pigs left the latter unaffected.

Parallel vaccination trials by this procedure, carried out in the laboratory, allowed us to follow the fate of the vaccine bacilli thus immobilised and protected. We were able to establish that they degrade rapidly, and that the resistance of calves so treated to the virulent experimental challenge disappears very quickly, as does their capacity to react to tuberculin.

We therefore abandoned this method at the end of December 1921, and decided to use, from 1 January 1922, the *intravenous* route, whose efficacy we had established with M. Calmette.¹

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III CAMPAIGN OF 1922

OLD TUBERCULOUS HOLDING AT 1 JANUARY 1922 · DIED OR SLAUGHTERED AND FOUND TUBERCULOUS IN 1921

At the beginning of 1922, the holding covered by the infection order of 18 July 1919 had been reduced to 10 head by the two slaughterings carried out in 1921, a cow and a heifer having been delivered in June and November to the private slaughterhouse of Contremoulins, where no apparent lesion was detected on inspection.

On 14 February 1922, the results given below of a general tuberculin testing of the holding (20 cows, 20 heifers, 19 bullocks and 5 calves, that is 64 head), carried out on 11 and 13 February by the combined methods of sub-caudal intradermal inoculation and the ophthalmic test, showed that the tuberculous infection was still persistent and severe. Reacted clearly to these tests:

- a* Ten cows, former reactors, remaining to be slaughtered from the holding declared infected in 1919 (no. 16, Boulette; 8, Hysope; 11, Idole; 28, Jannina; 30, Kava; 69, Javotte; 27, Kama; 33, Héroïne; 20, Kieva; 42, Darling).
- b* Three cows not reacting in 1919 (no. 4, Istrie; 56, Kabot; 13, Kamala).
- c* Six heifers born since 1919 (no. 17, Lucine; 7, Mica; 4, Mantille; 32, Lisière; 70, Mireille; 29, Manon).
- d* One bullock born since 1919 (no. 64).

Gave suspect reactions following this double test:

- e* Three cows not reacting in 1919 (no. 26, Kola; 21, Kermesse; 19, Kali).
- f* Six heifers born since 1919 (no. 18, Leda; 5, Mascotte; 79, Mirande; 71, Modane; 72, Mauviette; 31, Matelotte).

That is, 20 clear reactions and 9 suspect out of 64 cattle tested.

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The 7 males and the 10 heifers vaccinated in 1921 showed no reaction to these local tuberculin tests.

From January and through the course of the year, all newborn calves were to be vaccinated by a *single intravenous inoculation of 1 to 2 milligrammes* of BCG. We shall say later why the doses inoculated were so small.

¹ These *Annales*, September 1920.

NUMBER OF ANIMALS VACCINATED

There were, in 1922, 15 *births* (7 males and 8 females). All these animals were vaccinated by intravenous inoculation of BCG: 1 on 14 February; 4 on 12 April; 2 on 4 May; 4 on 21 June; 1 on 11 July; 2 on 5 September; 1 on 16 September. As to the age of the calves at the time of the intervention, the vaccination operations are distributed as follows:

AGE AT VACCINATION – 1922 GENERATION

1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH WEEK
0	2	1	0	4	3	1	1	1	2

No damaging change was observed in the condition or the growth of the vaccinated animals following the inoculation.

MORTALITY DURING THE YEAR AMONG THE VACCINATED, FROM CAUSES OTHER THAN TUBERCULOSIS

Only one death was observed in 1922, on 2 January: heifer *Ophir*, no. 53, born 3 March 1921, vaccinated on 15 March 1921, on the twelfth day, by subcutaneous insertion of a pellet, and died of congestion at ten months.

ANIMALS OF THE PRECEDING YEAR REVACCINATED BY INTRAVENOUS INOCULATION

7 males and 10 heifers remaining from the cohort born in 1921, which had originally been vaccinated with the pellet during that year, within the intervals indicated above, were subjected to *intravenous* revaccination on 14 February 1922, at the dose of 20 milligrammes.

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Parallel vaccination trials by this *intravenous* method, carried out in the laboratory on calves of one to four weeks, showed us that the dose of BCG which we believed must be effective (10 to 20 milligrammes) provoked in these very young animals, from the thirteenth day after inoculation, a violent bout of fever, a veritable crisis of typho-bacillosis, which lasted eight days and more and was accompanied by alarming general symptoms (loss of appetite and consequent wasting, arrest of growth). That is why we had to limit ourselves, in our Gruville trials, to the dose of 1 to 2 milligrammes, which seemed to us weak for the result we were expecting.

Furthermore, while this practice of intravenous inoculation is very easy to carry out on young animals, it becomes more delicate as the animals advance in age, their immobilisation becoming more and more difficult.

All these reasons decided us to adopt, from this year 1922 onward, the method of vaccination *by a single subcutaneous inoculation of 50 milligrammes of BCG, carried out within the first fifteen days after birth. This is the method we have used exclusively since 1 January 1923.*¹

IV

CAMPAIGN OF 1923

OLD TUBERCULOUS HOLDING AT 1 JANUARY 1923 · DIED OR SLAUGHTERED AND FOUND TUBERCULOUS IN 1922

At the beginning of 1923, the holding of 26 cattle covered by the infection order of 16 July 1919 had been reduced to five head by the five slaughterings carried out in 1922, following those of 1919, 1920 and 1921:²

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No. 33, *Héroïne*, cow of seven years: slaughtered 30 March 1922 at the Rouen abattoirs; fibro-calcareous tuberculosis of the thoracic organs and of the costal pleura. Condemnation of the lungs and the thoracic cage, 85 kilogrammes.

No. 27, *Kama*, cow of five and a half years: slaughtered 6 April 1922 at the Contremoulins slaughterhouse; tubercles, at various stages, in the left lung.

No. 20, *Kieva*, cow of five years: slaughtered 20 April 1922 at the Contremoulins slaughterhouse; tubercles, at various stages, in the lungs and on the costal pleura; condemnation of the thoracic organs and of the two forequarters: 139 kilos.

No. 16, *Boulette*, cow of eleven years: slaughtered 16 June 1922 at the Contremoulins slaughterhouse: calcified tubercles in the lung, the liver and the spleen.

No. 12, *Darling*, cow of eleven years: slaughtered 6 July 1922 at the Contremoulins slaughterhouse: large tubercles in the lungs.³

From January and through the following years, *all newborn calves were to be vaccinated by a single subcutaneous inoculation of BCG.*

NUMBER OF NEWBORN ANIMALS VACCINATED

There were 19 births in 1923, including two twin parturitions giving 4 males (14 males and 5 females). But, owing to the death of 3 males and 2 females which succumbed to septicaemia of the newborn or to infectious arthritis in the first month, and to the slaughter of another badly conformed male, the number of calves vaccinated in 1923 was reduced to 11 (9 *males* and 2 *females*). 2 calves born in the last days of 1923 count among the vaccinated of the 1924 campaign.

These vaccination operations, by subcutaneous inoculation into the dewlap of 50 milligrammes of BCG in 10 cubic centimetres of liquid excipient, were staged as follows, according to the course of the births: 6

¹ These *Annales*, May 1924.

² **TN** — The infection order is dated 16 July 1919 here and 18 July 1919 on p. 238; the order first introduced on p. 234 is that of 16 September 1918. The inconsistency is in the original.

³ **TN** — The cow numbered 12 here is called *Darling*; on p. 238 the same name is given as no. 42. One of the two is a misprint in the original.

February (1), 14 April (4), 8 March (1), 1 June (1), 9 June (1), 2 July (2), 26 October (1).¹ As to the age of the calves at initial vaccination, we note:

AGE AT VACCINATION – 1923 GENERATION

1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH WEEK
1	1	1	1	2	2	1	2

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ANIMALS OF THE TWO PRECEDING YEARS REVACCINATED BY THE SUBCUTANEOUS ROUTE IN 1923

On 6 February 1923 there were *revaccinated by inoculation under the skin of the dewlap* 7 males and 10 females remaining from the generation of 1921 (originally vaccinated in that year by the vaccine-ball procedure, and revaccinated by the intravenous route on 14 February 1922) and 7 males and 8 females of the generation of 1922 (originally vaccinated in that year by the intravenous route), that is a total of 32 revaccinations in 1923 (14 males and 18 females).

SEQUELAE OF THE VACCINAL INTERVENTION

No damaging change was observed in the general condition or the growth of the vaccinated of the 1923 generation, or of the revaccinated of the 1921 and 1922 generations, following subcutaneous inoculation of BCG. The only local effect of this inoculation is a painless induration, more or less adherent to the skin according to the depth of the injection, of variable volume and shape (walnut, pigeon's or hen's egg, apple, the size of a fist, or an elongated and diffuse thickening) persisting for some months to a year and more.

MORTALITY DURING THE YEAR FROM CAUSES OTHER THAN TUBERCULOSIS

Two males of the 1923 generation died: no. 47, born 19 January 1923, vaccinated on 6 February on the eighteenth day, died on 15 May at four months, of peritonitis from intestinal ulceration; no. 20, born 22 February 1923 (twin birth), vaccinated on 14 April on the fifty-first day, died on 21 April at two months, of infectious broncho-pneumonia of calves.

V CAMPAIGN OF 1924

OLD TUBERCULOUS HOLDING AT 1 JANUARY 1924 · DIED OR SLAUGHTERED AND FOUND TUBERCULOUS

No animal of the holding covered by the infection order of 16 July 1919 was delivered to the butcher in 1923, and the

¹ **TN** — The dates are printed out of chronological order, with 14 April preceding 8 March. The listed numbers total 11, matching the eleven calves vaccinated.

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5 reacting cows of 1919, which remained from that holding at the beginning of 1923, were still present at the beginning of 1924. The activity of the contagion remained manifest through two findings of pulmonary tuberculosis, made at the Contremoulins slaughterhouse by M. François Boissière senior, veterinary inspector: on 5 October 1923, in the cow *Lucine*, five years (no. 17), born at Gruville in 1918, not reacting to tuberculin in 1919 and not covered by the order of 16 July 1919, but reacting on 14 February 1922; on 31 January 1924, in 1 bullock of four years (no. 39), bought in September 1920, and which had not reacted to tuberculin on 14 February 1922.

NUMBER OF NEWBORN ANIMALS VACCINATED

There were 17 births in 1924 (10 males and 7 females). 6 of these calves were out of dams of the 1921 generation vaccinated in 1921, 1922, 1923 and 1924, and 2 were out of dams born at the beginning of 1922, vaccinated in 1922, 1923 and 1924.

Owing to the death of 4 males succumbing to dysentery and to infectious arthritis, and of one female whose autopsy revealed an intestinal perforation by a foreign body, the number of vaccinations made in 1924 on newborns was limited to 11 (5 males and 6 females): 1 male (no. 28), born 19 December 1924, counting among the vaccinated of 1925, just as 2 calves, born on 16 and 18 December 1923, are added to the vaccinated of 1924 (the operation being carried out on 17 January 1924). So, in total, 13 initial vaccinations in 1924 (6 males and 7 females). These interventions were staged as follows: 17 January (2), 1 April (1), 5 April (2), 15 April (2), 25 April (2), 5 June (1), 12 July (1), 17 September (1), 16 December (1). As to the age of the calves at initial vaccination, we note:

AGE AT VACCINATION — 1924 GENERATION

1ST	2ND	3RD	4TH	5TH WEEK
0	4	3	4	2

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Nota. — Owing to notification of births reaching us too late, it must be remarked that the majority of these vaccinations took place beyond the fifteen-day interval intended. The same remark applies to certain initial vaccinations of 1923.

ANIMALS OF THE THREE PRECEDING YEARS REVACCINATED IN 1924

On 19 February 1924, 43 cattle of the three preceding generations (of which 22 bullocks and 21 females) were re-inoculated with BCG by the subcutaneous route, namely: 7 bullocks and 10 females born in 1921, vaccinated successively with BCG by the ball in 1921, by the intravenous route in 1922 and by the subcutaneous route in 1923; 7 bullocks and 8 females born in 1922, vaccinated successively by the intravenous route in 1922 and by

the subcutaneous route in 1923; 8 bullocks and 3 females born in 1923, vaccinated initially by the subcutaneous route.

DIED DURING THE YEAR FROM CAUSES OTHER THAN TUBERCULOSIS

1 male (no. 3), born 13 March 1924, vaccinated on 5 April on the twenty-third day, died on 20 May, at two months and six days, of dysentery. One female (no. 64), *Robbia*, born 12 April 1924, vaccinated on 25 April on the thirteenth day, died on 27 July, at three and a half months, of an intestinal perforation.

No mortality occurred among the vaccinated of the 1921, 1922 and 1923 generations, whose general condition and growth have always been satisfactory. All these animals were found in fine condition at the general inspection made on 5 July 1924.

VACCINATED ANIMALS SLAUGHTERED FOR THE BUTCHER IN 1924 · AUTOPSIES

7 bullocks and 2 heifers, born in 1921, and 2 bullocks, born in 1922, were delivered to the butcher, slaughtered at the Contremoulins slaughterhouse and inspected by M. François Boissière:

No. 43, Orphée, female born 29 January 1921 (out of *Jannina* no. 28, reactor in 1919), vaccinated by the pumice-stone ball on 17 February 1921, on the nineteenth day; revaccinated on 14 February 1922 by the intravenous route and on 6 February 1923 and 19 February 1924 by the subcutaneous route. Heifer of fine conformation, fattened on account of sterility. Slaughtered on 9 April 1924, at three years two months and eleven days.

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No. 1, bullock, born 29 January 1921 (out of *Kamina* no. 59, not reacting in 1919), vaccinated 17 February 1921 (ball) on the nineteenth day; revaccinated 17 February 1922 by the intravenous route; on 6 February 1923 and 19 February 1924 by the subcutaneous route. Slaughtered 15 April 1924 at three years two months and eleven days.

No. 23, bullock, born 21 February 1921 (out of *Javotte* no. 69, reactor in 1919 and 1922), vaccinated 15 March 1921 (ball) on the twenty-second day; revaccinated 14 February 1922 (in the vein) and on 6 February 1923 and 19 February 1924 by the subcutaneous route. Delivered to the butcher on 13 May 1924 at three years two months and twenty-four days.

No. 15, bullock, born 26 February 1921 (out of *Kola* no. 25, not reacting in 1919, suspect reaction in 1922), vaccinated 15 March 1921 (ball) on the seventeenth day; revaccinated 14 February 1922 (in the vein) and on 6 February 1923 and 19 February 1924 by the subcutaneous route. Delivered to the butcher on 13 May 1924 at three years two months and nineteen days.

No. 58, bullock, born 3 March 1921 (out of *Floride* no. 22, not reacting in 1919), vaccinated 15 March 1921 (ball) on the twelfth day; revaccinated 14 February 1922 (in the vein) and on 6 February 1923 and 19 February 1924 by the subcutaneous route. Delivered to the butcher on 29 May 1924 at three years two months and twenty-six days.

No. 35, bullock, born 22 June 1921 (out of *Héroïne* no. 33, reactor in 1919 and 1922, slaughtered 30 December 1922; thoracic tuberculosis),¹ vaccinated 5 July 1921 (ball) on the thirteenth day; revaccinated 14 February 1922 (in the vein) and on 6 February 1923 and 19 February 1924 by the subcutaneous route. Delivered to the butcher on 15 September 1924 at three years two months and twenty-three days.

No. 40, bullock, born 23 March 1921 (out of *Kali* no. 19, not reacting in 1919), vaccinated 20 April 1921 (ball) on the twenty-eighth day; revaccinated 14 February 1922 (in the vein) and on 6 February 1923 and 19 February 1924 by the subcutaneous route. Delivered to the butcher on 17 September 1924 at three years five months and twenty-five days.

No. 45, Osborne, female born 28 March 1921 (out of *Kamala* no. 13, not reacting in 1919, reactor in 1922), vaccinated 20 April 1921 (ball) on the twenty-third day; revaccinated 14 February 1922 (in the vein) and on 6 February 1923 and 19 February 1924 by the subcutaneous route. Delivered to the butcher on 29 October 1924 at three years and seven months.

No. 14, bullock, born 8 October 1921 (out of *Lisière* no. 32, not reacting in 1919, reactor in 1922), vaccinated 20 October 1921 (ball) on the twelfth day; revaccinated 14 February 1922 (in the vein) and on 6 February 1923 and 19 February 1924 by the subcutaneous route. Delivered to the butcher on 31 December 1924 at three years two months and twenty-three days.

No. 77, bullock, born 24 February 1922 (out of *Mirande* no. 79, suspect reaction in 1922), vaccinated 12 April 1922 (in the vein) on the forty-seventh day; revaccinated on 6 February 1923 and 19 February 1924 by the subcutaneous route. Delivered to the butcher on 31 December 1924 at two years ten months and seven days.

No. 42, bullock, born 8 March 1922 (out of *Matelotte* no. 31, suspect reaction in 1922), vaccinated 12 April 1922 (in the vein) on the thirty-fifth day; revaccinated on 6 February 1923 and 19 February 1924 by the subcutaneous route. Delivered to the butcher on 31 December 1924 at two years nine months and twenty-three days.

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The slaughter of these 11 animals gave rise to no finding of tuberculosis.

VI CAMPAIGN OF 1925

OLD TUBERCULOUS HOLDING AT 1 JANUARY 1925 · DIED OR SLAUGHTERED AND FOUND TUBERCULOUS

At the beginning of 1925, the holding of 26 cattle covered by the infection order of 16 July 1919 had been reduced to two head by the three slaughterings carried out at the Contremoulins slaughterhouse in 1924, following those of the preceding years:

No. 28, Jannina, cow of seven and a half years, slaughtered 24 January 1924 (hepatic distomatosis and echinococcosis, without apparent lesions of tuberculosis).

No. 8, Hysope, cow of eight and a half years, slaughtered 17 January 1924. Some tubercles in the right lung.

No. 30, Kava, cow of seven years, slaughtered 16 November 1924 (old lesions of tuberculosis of both lungs).

¹ **TN** — No. 33 *Héroïne* is here given as slaughtered on 30 December 1922; on p. 241 her slaughter is dated 30 March 1922 at Rouen. The discrepancy is in the original.

NUMBER OF NEWBORN ANIMALS VACCINATED IN 1925

There were 25 births in 1925 (10 males and 15 females). 9 of these calves were out of vaccinated dams of the 1921 generation, one of which, *Obéron*, gave 2 calves in 1925 (1 female on 18 January and 1 male on 28 December), and seven were out of vaccinated dams of the 1922 generation.

Owing to the death of 2 females which succumbed to dysentery in the first week, the number of vaccinations made in 1925 on newborns was limited to 23 (10 males and 13 females). 1 calf, a male (no. 82) born 28 December 1925, counts among the vaccinated of 1926, just as a male (no. 28), born 19 December 1924, is added to those vaccinated in 1925 (the operation being carried out on 8 January 1925, on the twentieth day), that is a total of 23 initial vaccinations in 1925.

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The vaccination operations were staged as follows: 8 January (1), 15 January (3), 24 January (2), 21 February (2), 17 March (3), 29 March (1), 2 June (2), 26 June (1), 13 August (1), 1 September (1), 4 September (1), 12 November (1), 23 October (1), 7 December (1), 10 December (1), 23 December (1).¹ As to the age of the calves at initial vaccination, we note:

AGE AT VACCINATION – 1925 GENERATION

1ST WEEK	2ND WEEK	3RD WEEK	6TH WEEK
12	9	1	1

That is 21 vaccinations within the interval fixed by the technique of the method, and 2 late operations (no. 28, male, born 19 December 1924 and vaccinated 8 January 1925, on the twentieth day) and no. 84 (*Soucieuse*, female born 8 July and vaccinated 12 August, on the thirty-sixth day).

ANIMALS OF THE FOUR PRECEDING YEARS REVACCINATED

On 21 February 1925, 42 cattle of the four preceding generations (18 bullocks and 24 females) were re-inoculated by the subcutaneous route, namely: 8 females remaining from the generation originally vaccinated in 1921; 5 bullocks and 8 females remaining from the generation originally vaccinated in 1922; 8 bullocks and 3 females remaining from the generation originally vaccinated in 1923; 5 bullocks and 5 females remaining from the generation originally vaccinated in 1924.

MORTALITY DURING THE YEAR FROM CAUSES OTHER THAN TUBERCULOSIS

Male, born 14 January 1925, vaccinated on the 25th, on the tenth day, died 21 April 1925, at three months and seven days, of infectious broncho-pneumonia.

¹ **TN** – The dates are printed with 12 November preceding 23 October. The listed numbers total 23, and the age distribution likewise totals 23, matching the twenty-three initial vaccinations of 1925.

Male, no. 30, born 20 February 1926, vaccinated on 21 February, died on 31 October: broncho-pneumonia.¹

Signora, no. 82, female, born 13 March 1925, vaccinated on the 17th on the fourth day, died 27 June 1925, at three months and fourteen days, of broncho-pneumonia.

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Sévère, no. 75, female, born 25 March 1925, vaccinated on the 28th on the fourth day, died on 4 April, at ten days, of septicaemia.

Male, no. 70, born 9 March 1925, vaccinated 17 March on the eighth day, died on 25 October of broncho-pneumonia, at seven months and sixteen days.

Sibérie, no. 83, female, born 20 June 1925, vaccinated on the 26th on the sixth day, died on 28 September, at three months and six days, of broncho-pneumonia.

Furthermore, it was necessary to deliver to the butcher, on account of poor conformation, the male no. 40, born 14 January 1925, vaccinated on the 24th, on the tenth day, slaughtered on 5 March, at two months and three weeks.

Among the vaccinated of the 1924 generation, revaccinated in 1925, a male, no. 8, born 21 March 1924, died on 6 July 1925, at fifteen and a half months, of verminous bronchitis, and no. 77, *Rosa*, female, born 8 December 1924, succumbed to symptomatic anthrax [blackleg] on 4 October 1925, at ten months.

VACCINATED ANIMALS SLAUGHTERED FOR THE BUTCHER IN 1925 · AUTOPSIES

One female, born in 1921, and 4 bullocks, born in 1922, were sent to the Contremoulins slaughterhouse in 1925:

No. 68, *Ortie*, female, born 12 February 1921 (out of *Kieva*, reactor of 1919, slaughtered in 1922 and found to have thoracic tuberculosis, herself out of *Fanchon*, slaughtered in 1920 and likewise affected with thoracic tuberculosis). Vaccinated 3 March 1921 (ball); revaccination on 14 February 1922 (in the vein) and on 6 February 1923, 19 February 1924 and 21 February 1925 by the subcutaneous route. Delivered to the butcher on 5 March 1925, at four years two months and twenty-one days.

No. 36, bullock, born 3 February 1922 (out of *Kabot*, no. 56, not reacting in 1919, reactor in 1922), vaccinated 14 February 1922 (in the vein) on the eleventh day; revaccinated on 6 February 1923, 19 February 1924 and 21 February 1925 by the subcutaneous route. Delivered to the butcher on 11 March 1925, at three years one month eight days.

No. 22, bullock, born 4 March 1922 (out of *Idole*, no. 11, reactor in 1919), vaccinated 12 April 1922 (in the vein) on the thirty-ninth day; revaccinated on 6 February 1923, 19 February 1924 and 21 February 1925 by the subcutaneous route. Delivered to the butcher on 11 March 1925, at three years seven days.

No. 39, bullock, born 4 April 1922 (out of *Javotte*, no. 69, reactor in 1919), vaccinated 4 March 1922 (in the vein), at one month;² revaccinated on 6 February 1923, 19 February 1924 and 21 February 1925, by the subcutaneous route. Delivered to the butcher on 15 April 1925.

¹ **TN** — This entry is dated 1926 in a list of 1925 mortality, and the number 82 is used both for the female *Signora* and for the male calf born 28 December 1925 on p. 246. Both inconsistencies are in the original.

² **TN** — No. 39 is given as born 4 April 1922 and vaccinated 4 March 1922, a month before birth; one of the two dates is a misprint in the original. A bullock no. 39 also appears on p. 243 as a four-year-old bought in September 1920 and found tuberculous in January 1924, so the number is reused.

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No. 66, bullock, born 25 April 1922 (out of *Mantille*, no. 4, born in 1919, reactor in 1922), vaccinated 21 June 1922 (in the vein), on the fifty-seventh day; revaccinated on 6 February 1923, 19 February 1924 and 21 February 1925, by the subcutaneous route. Delivered to the butcher on 5 May 1925 at three years ten days.

The slaughter of these five animals gave rise to no finding of tuberculosis.

VII

CAMPAIGN OF 1926

OLD TUBERCULOUS HOLDING AT 1 JANUARY 1926 · DIED OR SLAUGHTERED AND FOUND TUBERCULOUS

At the beginning of 1926 the old holding covered by the infection order of 16 July 1919 was still represented by two cows remaining at 1 January 1925: *Idole*, no. 11, nine and a half years, and *Javotte*, no. 69, nine years.

Another cow of the farm, *Modane*, no. 71, born in 1919, consequently not vaccinated and not covered by the infection order, had given a doubtful ophthalmo-reaction in 1922 and a positive intradermo-reaction in 1923. Slaughtered on 4 November 1926, in excellent condition of fatness, at the Contremoulins slaughterhouse, she was condemned in her entirety for *generalised tuberculosis*. This finding shows the persistence of contagion at Gruville, manifesting itself in severe cases in unvaccinated animals.

NUMBER OF NEWBORN ANIMALS VACCINATED

There were 26 births in 1926 (12 males and 14 females). 6 of these calves are out of dams of the generation originally vaccinated in 1921; 8 are out of dams of the generation originally vaccinated in 1922; and 2 are out of dams of the generation originally vaccinated in 1923.

ANIMALS OF THE FIVE PRECEDING YEARS REVACCINATED

On 6 March 1926, 42 cattle remaining from the five preceding generations (10 bullocks and 32 females) were revaccinated by the subcutaneous route, namely:

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7 females remaining from the generation of those originally vaccinated in 1921; 8 females remaining from the generation of those originally vaccinated in 1922; 1 bullock and 3 females remaining from the generation of those originally vaccinated in 1923; 4 bullocks and 4 females remaining from the generation of those originally vaccinated in 1924; 5 bullocks and 10 females remaining from the generation of those originally vaccinated in 1925.

MORTALITY DURING THE YEAR FROM CAUSES OTHER THAN TUBERCULOSIS

No. 72, Thésée, female, born 26 January 1926, vaccinated on 3 February, on the eighth day, died on 7 December 1926 of indigestion and obstruction of the omasum, at ten and a half months.

No. 3, Thaïs, female, born 2 May 1926, vaccinated on 10 May, on the eighth day, died on 10 July 1926 of pulmonary congestion, at two and a half months.

No. 24, Thémis, female, born 4 May 1926, vaccinated on 10 May, on the sixth day, died on 18 September 1926 from intestinal torsion, at four and a half months.

A male calf, born 17 December 1925, no. 70, vaccinated on 25 December 1925, on the eighth day, had to be slaughtered on 27 January 1926, at forty days, on account of emphysema of the head and neck.

VACCINATED ANIMALS SLAUGHTERED FOR THE BUTCHER · AUTOPSY

14 vaccinated cattle (12 bullocks and 2 females) were delivered to the butcher in 1926, namely: 1 bullock, born in 1922; 9 bullocks and 2 females, born in 1923; 2 bullocks, born in 1924.

No. 48, bullock, born 30 April 1923 (out of *Matelotte*, not reacting in 1919), vaccinated 1 June 1923 on the thirty-second day; revaccinated on 19 February 1924 and 21 February 1925. Delivered to the butcher at Theuville-aux-Maillots (Seine-Inférieure) on 5 February 1926, at two years ten months twenty-five days.

No. 20, bullock, born 7 June 1923 (out of *Kava*, reactor in 1919 and 1922, slaughtered on 16 November 1924 and found to have pulmonary tuberculosis), vaccinated 9 June 1923 on the second day, by the subcutaneous route; revaccinated on 19 February 1924 and 21 February 1925. Delivered to the butcher at Theuville-aux-Maillots, on 5 February 1926, at two years seven months twenty-eight days.

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No. 72, bullock, born 21 June 1923 (born twin out of *Mirande*, not reacting in 1919), vaccinated 2 July 1923, on the eleventh day, by the subcutaneous route; revaccinated on 19 February 1924 and 21 February 1925. Delivered to the butcher at Theuville-aux-Maillots, on 5 February 1926, at two years seven months fourteen days.

No. 27, bullock, born 8 September 1922 (out of *Hysope*, reactor in 1919), vaccinated 19 September 1922 (*in the vein*), on the eighteenth day; revaccinated by the subcutaneous route on 6 February 1923, 19 February 1924 and 21 February 1925. Delivered to the butcher at Contremoulins, on 5 March 1926, at three years five months twenty-five days.

No. 36, bullock, born 13 March 1923 (out of *Lisière*, not reacting in 1919), vaccinated 14 April 1923, on the thirty-third day, by the subcutaneous route; revaccinated on 19 February 1924 and 21 February 1925. Delivered to the butcher at Contremoulins, on 5 March 1926, at two years eleven months twenty-three days.

No. 12, bullock, born 19 February 1923 (out of *Lucine*, not reacting in 1919, reactor in 1922, slaughtered 5 October 1923 and found to have pulmonary tuberculosis), vaccinated 14 April 1923, by the subcutaneous route, on the forty-fourth day; revaccinated on 19 February 1924 and 21 February 1925. Delivered to the butcher at Contremoulins, on 5 March 1926, at three years fourteen days.

No. 24, bullock, born 12 September 1923 (out of *Mantille*, not reacting in 1919), vaccinated 26 October 1923, on the forty-fourth day, by the subcutaneous route; revaccinated on 19 February 1924 and 24 February 1925. Delivered to the butcher at Contremoulins, on 5 March 1926, at two years five months twenty-three days.

No. 73, bullock, born 21 June 1923 (born twin out of *Mirande*, mentioned above), vaccinated 2 July 1923, on the eleventh day, by the subcutaneous route; revaccinated on 19 February 1924 and 21 February 1925. Delivered to the butcher at Contremoulins, on 5 March 1926, at two years eight months twelve days.

No. 29, Quina, female, born 1 March 1923 (out of *Kola*, not reacting in 1919), vaccinated 14 March 1923, on the thirteenth day, by the subcutaneous route; revaccinated on 19 February 1924, 21 February 1925 and 6 March 1926. Delivered on account of sterility to the butcher at Contremoulins, on 31 March 1926, at three years twenty-eight days.

No. 74, Quita, female, born 8 May 1923 (out of *Javotte*, reactor of 1919), vaccinated 1 June 1923, on the twenty-fourth day, by the subcutaneous route; revaccinated on 19 February 1924, 21 February 1925 and 6 March 1926. Delivered to the butcher at Contremoulins, on 23 September 1926, at three years four months fifteen days.

No. 17, bullock, born 16 December 1923 (out of *Léda*, not reacting in 1919), vaccinated 17 January 1924, on the thirty-second day, by the subcutaneous route; revaccinated on 21 February 1925 and 6 March 1926. Delivered to the butcher at Contremoulins, on 7 October 1926.

No. 2, bullock, born 12 March 1924 (out of *Othello*, vaccinated since 1921), vaccinated 5 April 1924, on the twenty-fourth day, by the subcutaneous route; revaccinated on 21 February 1925 and 6 March 1926. Delivered to the butcher at Contremoulins, on 14 October 1926, at two years seven months.

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No. 6, bullock, born 19 March 1924 (out of *Ophile*, vaccinated since 1921), vaccinated 5 April 1924, on the sixteenth day, by the subcutaneous route; revaccinated on 21 February 1925 and 6 March 1926. Delivered to the butcher at Contremoulins, on 21 October 1926, at two years seven months two days. Live weight: 521 kilogrammes. Gave 252 kilogrammes of meat, that is 48.36 per 100 dressing yield.

No. 1, bullock, born 2 July 1924 (out of *Mirande* no. 79, suspect reaction in 1922), vaccinated 12 July 1924, by the subcutaneous route, on the tenth day; revaccinated on 21 February 1925 and 6 March 1926. Delivered to the butcher on 4 November 1926, at two years four months two days. Live weight: 580 kilogrammes. Gave 276 kilogrammes of meat, that is 47.51 per 100 dressing yield.

The slaughter of these 14 animals gave rise to no finding of tuberculosis.

The recapitulative table below shows the course of the operations and the situation of the Gruville farm at 1 January 1927. The number of tuberculous animals and of those which, born before 1921 and left in the infected environment, must consequently be eliminated, has fallen from 64 head in 1921 to 11 head in 1927, in accordance with economic necessity. In accordance with our wishes, the tuberculous infection has remained permanent in the course of these six years; it is still represented *officially* by the last two cows covered by the prefectural order of 16 July 1919.

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VIII CONCLUSIONS

Independently of the 58 cattle, vaccinated and revaccinated, which currently populate the byres of Gruville, and which have successively taken the place, at minimum cost, of the old tuberculous herd, 30 vaccinated and revaccinated cattle (25 bullocks and 5 cows) born and reared on this infected holding have reached, in the course of the experiment, the end of their economic career, that is to say the abattoir, without having contracted tuberculosis. Among these last 30 animals, 20 were born of tuberculous or suspect dams, and had

RECAPITULATIVE TABLE — COURSE OF THE OPERATIONS, 1921 TO 1926

YEAR	OLD TUBERCULOUS OR CONTAMINATED HOLDING (TO BE ELIMINATED)		NEW VACCINATED AND REVACCINATED HOLDING			
	TUBERCULOUS STOCK AND ANIMALS BORN BEFORE 1921, LEFT IN THE INFECTED ENVIRONMENT	DIED OR SLAUGHTERED AND FOUND TUBERCULOUS	ANIMALS VACCINATED AT BIRTH	VACCINATED ANIMALS DIED OF CAUSES OTHER THAN TUBERCULOSIS	VACCINATED AND REVACCINATED ANIMALS SENT TO THE ABATTOIR AT THE END OF THEIR CAREER	FOUND TUBERCULOUS
1921	64	0	22	4	—	—
1922	46	5	15	1	—	—
1923	34	1	11	2	—	—
1924	20	3	13	2	11	0
1925	15	0	23	9	5	0
1926	12	1	26	4	14	0

Situation at 1 January 1927: 11 cows remain to be eliminated; 58 vaccinated and revaccinated cattle have taken the place of the old herd. **TN** — In the original the two groups of columns are headed *Ancien effectif tuberculeux ou contaminé (à éliminer)* and *Nouvel effectif vacciné et revacciné*; the em dashes stand for the original's ditto marks, meaning not applicable.

been fed, during their first months, with the mother's milk, unboiled. We have said, furthermore, that all measures of isolation and disinfection had been suspended.

It remains only for us to recall the three qualities that a method of prophylaxis must possess in order to enter into current practice, namely: *simplicity, harmlessness, efficacy*.

It is needless to insist on the simplicity of an intervention consisting of a subcutaneous inoculation renewed each year.

The harmlessness of BCG is superabundantly demonstrated by the fact that newborn animals, vaccinated in 1921, then revaccinated over three and four consecutive years, were found at slaughter free of any tuberculous lesion. We are entitled to conclude that BCG is incapable of becoming virulent again, even in the organism of origin, that of the ox.

Furthermore, the normal growth and the fattening of the vaccinated and revaccinated animals have in no way been affected, since the butchery yields of the bullocks slaughtered have been 47 to 53 per 100 (Normandy method).

As to efficacy, the results of the experiment, which we have just related, allow the problem posed at the outset of these trials, as we stated it above, to be considered as practically resolved.

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TRANSLATION

Complete English translation of the paper published in the Annales de l'Institut Pasteur, 41^e année, n° 3, March 1927, pp. 233–253. The article immediately following in the same issue, at p. 254, is B. Weill-Hallé and R. Turpin on BCG vaccination of children.